MAT 017B Quiz 7

September 6, 2018

Name: Key

SID:

1. A biologist wants to monitor the activity of three species of mollusk in an underwater cave. The biologist sets up different sensors in the cave. There is a sensor to detect arms, eyes, and shells. Individuals of species X have 2 arms and 2 eyes, individuals of species Y have 1 arm and 2 eyes, and individuals of species Z have 1 arm and 1 eye. Every species has 1 shell. After an hour, the sensors detected 5 arms, 7 eyes, and 4 shells. How many individuals of each species were detected by the sensors?

x – number of species X　　arms: $2x + y + z = 5$

y – number of species Y　　eyes: $2x + 2y + z = 7$

z – number of species Z　　shells: $x + y + z = 4$

$$\left[\begin{array}{ccc|c} 2 & 1 & 1 & 5 \\ 2 & 2 & 1 & 7 \\ 1 & 1 & 1 & 4 \end{array}\right] \xrightarrow[R_1 - 2R_3 \to R_3]{R_1 - R_2 \to R_2} \left[\begin{array}{ccc|c} 2 & 1 & 1 & 5 \\ 0 & -1 & 0 & -2 \\ 0 & -1 & -1 & -3 \end{array}\right]$$

$$\xrightarrow{R_2 - R_3 \to R_3} \left[\begin{array}{ccc|c} 2 & 1 & 1 & 5 \\ 0 & -1 & 0 & -2 \\ 0 & 0 & -1 & -1 \end{array}\right] \Rightarrow \begin{array}{l} x = \frac{1}{2}(5 - 1 - 2) = 1 \\ y = 2 \\ z = 1 \end{array}$$

$$\boxed{\begin{array}{l} 1 \text{ of species } 1 \\ 2 \text{ of species } 2 \\ 1 \text{ of species } 3 \end{array}}$$